TABLE S10

CONVENTIONAL B004 ANNOTATION CLASSIFIERS

Clustering	Classifier	Input	Agreement (%)	Macro-F1
Leiden	Nearest centroid	All means	58.20	0.4341
Leiden	Nearest centroid	Means + fractions	55.74	0.4370
Leiden	Logistic regression	All means	67.05	0.4981
Leiden	Logistic regression	Means + fractions	68.20	0.5030
FlowSOM	Nearest centroid	All means	35.63	0.2745
FlowSOM	Nearest centroid	Means + fractions	35.19	0.2787
FlowSOM	Logistic regression	All means	45.69	0.3078
FlowSOM	Logistic regression	Means + fractions	45.72	0.3059
SpatialSort	Nearest centroid	All means	36.10	0.1798
SpatialSort	Nearest centroid	Means + fractions	35.26	0.1778
SpatialSort	Logistic regression	All means	44.12	0.2165
SpatialSort	Logistic regression	Means + fractions	44.09	0.2170
PIXIE	Nearest centroid	All means	46.76	0.2551
PIXIE	Nearest centroid	Means + fractions	46.96	0.2577
PIXIE	Logistic regression	All means	49.95	0.2684
PIXIE	Logistic regression	Means + fractions	51.92	0.2728

Classifiers used B004 training-region cell labels, richer supervision than the LLM prompts. Hyperparameters were chosen in validation regions. Metrics average the two test regions equally; fixed 28-label macro-F1 includes absent classes as zero.